# CS 40700 Sprint 2 Planning

Product Name: Yarrow

Team Number: 13

Team:
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### Sprint Overview

The focus of Sprint 2 is to extend Yarrow by adding additional features. This includes strengthening document processing, enhancing extracted content display, and providing additional utilities that improves user experience.

By the end of this sprint, users should be able to upload and manage document batches with resumable transfers and progress tracking, receive quality warnings (blur, skew, truncation, brightness), export tables as CSV, manually correct extracted text, perform exact-phrase and OCR-tolerant searches with sorting, and manage administrative accounts and system monitoring.

**Scrum Master:** Cheng-En Lee

**Meeting Schedule:** Monday 11:00 - 11:30 AM

## Current Sprint Detail

### Document Upload & Batch Management

**User Story 80: Resumable File Uploads**
Owner: Anthony McCrovitz | Total Hours: 14 hrs
Description: As a user, I want to pause and resume interrupted uploads of large documents so that I do not have to restart the transfer from the beginning.

| Task | Estimated Hours |
|---|---|
| Implement resumable upload-session endpoints to create, inspect, and finalize uploads | 3 hrs |
| Write file chunks to temporary storage with size, checksum, and quota validation | 3 hrs |
| Add browser upload progress, retry, and resume controls for interrupted | 3 hrs |

| | |
|---|---|
| transfers | |
| Finalize assembled files safely, enqueue processing once, and clean up abandoned sessions | 2 hrs |
| Unit and integration testing for dropped connections, repeat chunks, quotas, and recovery | 3 hrs |

Acceptance Criteria:

- Given a large document upload is interrupted, when I reconnect and resume within the supported retention window, then the transfer continues from the last confirmed chunk instead of uploading the entire file again.
- Given a chunk is retried after a network timeout, when the server has already accepted it, then the server handles the retry without duplicating file content.
- Given all chunks are uploaded, when the upload is finalized, then one complete document is stored and exactly one processing job is queued.
- Given an upload would exceed my storage quota or the uploaded content fails validation, when the server checks the upload, then the request is rejected and the user receives a clear error.

**User Story 31: Batch Upload and Processing Progress**

Owner: Anthony McCrovitz | Total Hours: 9 hrs

Description: As a user, I want to see the overall status of documents submitted together so that I can track long-running batches without opening each document individually.

| **Task** | **Estimated Hours** |
|---|---|
| Associate files uploaded together with a batch identifier and stored membership | 2 hrs |
| Implement a user-authorized endpoint that aggregates queued, processing, completed, failed, and canceled counts | 2 hrs |
| Create batch progress panel on the Upload page and refresh statuses while the app remains open | 3 hrs |
| Unit and integration testing for independent document outcomes and progress updates | 2 hrs |

Acceptance Criteria:

- Given I upload multiple documents together, when the batch is created, then the application gives them a shared batch identifier without combining their individual processing jobs.
- Given a batch contains documents in different states, when I open its progress view, then I see accurate counts for queued, processing, completed, failed, and canceled documents.
- Given one document fails processing, when other documents finish, then the batch progress reflects each outcome without incorrectly failing the remaining documents.
- Given I leave the Upload page and return during the session, when batch data is fetched again, then I can see its latest persisted progress.

**User Story 79: Detect Duplicate Document Uploads**

Owner: Anthony McCrovitz | Total Hours: 7 hrs

Description: As a user, I want to be warned when I upload a document that is identical to one already in my library so that I can avoid unnecessary duplicate files.

| **Task** | **Estimated Hours** |
|---|---|
| Store a content fingerprint for completed uploads, scoped to each user's | 2 hrs |

| | |
|---|---|
| documents | |
| Add duplicate detection and an upload confirmation to skip or upload another copy | 3 hrs |
| Unit and integration testing for identical files, renamed files, different owners, and concurrent uploads | 2 hrs |

Acceptance Criteria:

- Given an identical document is already in my library, when I attempt to upload the same file, then the application warns me before accepting another copy.
- Given a duplicate warning, when I choose Skip, then the system does not create another document or consume additional storage quota.
- Given a duplicate warning, when I explicitly choose Upload Anyway, then the application accepts a separate document subject to normal size and quota rules.
- Given another user owns an identical file, when I upload mine, then the system does not disclose that other user's document or filename.

**User Story 60: Batch Completion Notifications**

Owner: Anthony McCrovitz | Total Hours: 6 hrs

Description: As a user, I want to be notified when a long-running batch finishes so that I can return to review the results instead of continuously checking progress.

| **Task** | **Estimated Hours** |
|---|---|
| Detect and persist a one-time transition when a batch reaches a terminal state | 1.5 hrs |
| Show a completion summary through an in-app notification while the user is signed in | 2 hrs |
| Optionally display a browser notification when permission is granted and prevent repeated alerts | 1 hr |
| Unit and frontend testing for success, mixed outcomes, and duplicate-notification prevention | 1.5 hrs |

Acceptance Criteria:

- Given a batch is still processing, when I navigate to another Yarrow page, then I can continue using the application without blocking on the batch.
- Given every document in my batch has reached a terminal state, when batch completion is detected, then I receive an in-app notification summarizing successful and failed documents.
- Given browser notifications are enabled, when the batch completes while Yarrow remains open, then a browser notification may be shown without requiring me to stay on the Upload page.
- Given I have already been notified about a completed batch, when the status is checked again, then the same completion event does not generate repeated notifications.

**User Story 81: Folder Upload Support**

Owner: Anthony McCrovitz | Total Hours: 4 hrs

Description: As a user, I want to choose or drag in a folder of supported documents so that I can submit multiple files without selecting each one manually.

| **Task** | **Estimated Hours** |
|---|---|
| Support directory selection and recursive file discovery in compatible | 1.5 hrs |

| | |
|---|---|
| browsers | |
| Validate discovered files and display a pre-upload file list with unsupported types identified | 1 hr |
| Connect the folder selection to the existing bulk-upload and batch workflows | 0.5 hrs |
| Frontend tests for nested folders, mixed file types, and empty folders | 1 hr |

Acceptance Criteria:

- Given I select a folder containing supported documents, when the file list is prepared, then Yarrow includes the documents from nested subfolders in the pending upload list.
- Given a selected folder includes unsupported or invalid files, when validation runs, then those files are identified without preventing supported files from being selected.
- Given I confirm a folder upload, when files are submitted, then the existing bulk-upload flow handles each accepted file and adds it to one batch.
- Given directory selection is not supported by the browser, when I open the uploader, then ordinary file selection remains usable.

## Document Processing Warnings

**User Story 32: Blurred Warning**
Owner: Cheng-En Lee | Total Hours: 6 hrs
Description: As a user, I would like to receive warnings that my document may be blurred instead of silently receiving poor extraction results.

| **Task** | **Estimated Hours** |
|---|---|
| Add blur detection to the worker nodes | 2 hrs |
| Implement UI to display warnings | 2 hrs |
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given a blurred image that is hardly recognizable by eye, when the user uploads the image, the system correctly warns the user about the blurred page.
- Given the warning, when the user clicks the link in the warning, the user should be navigated to the page that caused the warning.
- Given a clear image, when the user uploads the image, the system would not warn the user for the image being blurred.

**User Story 33: Skew Warning**
Owner: Cheng-En Lee | Total Hours: 6 hrs
Description: As a user, I would like to receive warnings that my document may be skewed instead of silently receiving poor extraction results.

| **Task** | **Estimated Hours** |
|---|---|
| Add skew detection to the worker nodes | 2 hrs |
| Implement UI to display warnings | 2 hrs |

| Task | Estimated Hours |
|---|---|
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given an image that is rotated significantly (eg. 45 degrees), when the user uploads the image, the system correctly warns the user about the skewed page.
- Given the warning, when the user clicks the link in the warning, the user should be navigated to the page that caused the warning.
- Given a properly aligned image, when the user uploads the image, the system would not warn the user for the image being skewed.

**User Story 34: Truncation Warning**

Owner: Cheng-En Lee | Total Hours: 6 hrs

Description: As a user, I would like to receive warnings that my document may be cropped/truncated instead of silently receiving poor extraction results.

| Task | Estimated Hours |
|---|---|
| Add truncation detection to the worker nodes | 2 hrs |
| Implement UI to display warnings | 2 hrs |
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given an image that has text cropped off, when the user uploads the image, the system correctly warns the user about the page being potentially cropped.
- Given the warning, when the user clicks the link in the warning, the user should be navigated to the page that caused the warning.
- Given an image with well-displayed text content, when the user uploads the image, the system would not warn the user for the image being potentially cropped.

**User Story 35: Brightness Warning**

Owner: Cheng-En Lee | Total Hours: 6 hrs

Description: As a user, I would like to receive warnings that my document may be too dark/too light instead of silently receiving poor extraction results.

| Task | Estimated Hours |
|---|---|
| Add brightness threshold to the worker nodes so overly dark/light frames can be identified | 2 hrs |
| Implement UI to display warnings | 2 hrs |
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given an image that is overly dark or overly light, when the user uploads the image, the system correctly raises a brightness warning for the page.
- Given the warning, when the user clicks the link in the warning, the user should be navigated to the page that caused the warning.

- Given an image with a brightness that is easily readable, when the user uploads the image, the system would not raise a brightness warning for the page.

## Document Processing Enhancements

**User Story 49: Export Tables as CSV**
Owner: Aagam Bapna | Total Hours: 5 hrs
Description: As a user, I want to export extracted tables as CSV so I can open them in a spreadsheet.

| Task | Estimated Hours |
|---|---|
| Build CSV from extracted tables | 1.5 hrs |
| Include every table in the download | 1 hr |
| Add a CSV option to the export menu | 1hr |
| Unit and integration testing | 1.5 hrs |

Acceptance Criteria:

- Given a processed document has tables, when I export CSV, then I get a `.csv` file.
- Given a document has more than one table, when I export, then every table is in the file.
- Given a document is still processing, when I try to export, then the export is blocked.

**User Story 92: Prioritize Small Jobs**
Owner: Aagam Bapna | Total Hours: 5 hrs
Description: As a user, I want small jobs to run sooner so a large batch does not block them.

| Task | Estimated Hours |
|---|---|
| Record job size for queue order | 1.5 hrs |
| Let small jobs go ahead of large batches | 1.5 hr |
| Keep large jobs moving so they are not stuck forever | 1hr |
| Unit and integration testing | 1 hr |

Acceptance Criteria:

- Given a large batch is already queued, when I upload a small document, then it is not placed behind the whole batch.
- Given two small jobs are queued, when they wait, then the earlier one stays ahead of the later one.
- Given a large job is already running, when a small job arrives, then the running job is not cancelled.

**User Story 45: Rotation Correction**
Owner: Daniel Benes-Magana | Total Hours: 8 hrs
Description: As a user, I would like rotated pages to be detected and corrected before OCR.

| Task | Estimated Hours |
|---|---|
| Implement orientation detection model/algorithm | 3 hrs |
| Auto-rotate image before passing to OCR engine | 3 hrs |

| Task | Estimated Hours |
|---|---|
| Integration testing | 2 hrs |

**User Story 46: Confidence Scores**
Owner: Cheng-En Lee | Total Hours: 4 hrs
Description: As a user, I would like extracted text to include a confidence score.

| Task | Estimated Hours |
|---|---|
| Extract confidence scores from OCR model output | 1 hr |
| Store and serve confidence scores via backend API | 1 hr |
| Display confidence indicators in text viewer UI | 2 hrs |

Acceptance Criteria:

- Given a document being processed, when the user selects each extracted content, the confidence score for the extracted content should be displayed to the user.
- Given extracted text has a low confidence score below the threshold, when the document is viewed, then the low-confidence region is visually highlighted for user review.
- Given a user inspects the document details or metadata, when confidence scores are aggregated, then an overall document confidence summary is available.

**User Story 51: Manual Correction**
Owner: Daniel Benes-Magana | Total Hours: 8 hrs
Description: As a user, I would like to manually correct incorrectly extracted text.

| Task | Estimated Hours |
|---|---|
| Build inline text editing interface in UI | 3 hrs |
| Create backend API endpoints to update extracted text | 3 hrs |
| Integration testing | 2 hrs |

**User Story 100: See Differences and Undo Corrections**
Owner: Daniel Benes-Magana | Total Hours: 6 hrs
Description: As a user, I would like to see the difference between the original extraction and my correction, and undo a correction.

| Task | Estimated Hours |
|---|---|
| Store original OCR text alongside revisions and build reversion endpoints | 2.5 hrs |
| Build inline diff modal/popover highlighting additions and deletions in the viewer | 2 hrs |
| Unit and integration testing for reverting corrections and diff rendering | 1.5 hrs |

Acceptance Criteria:

- Given a text region has been edited, when I view the region in the viewer, then I can see a visual diff comparing the original extraction against the corrected text.

- Given a corrected text region, when I choose to undo the correction, then the text is restored to the raw OCR output without data loss.
- Given an undone correction, when the document is viewed or searched, then the active text reverts to the original extraction immediately.

## Search Enhancements

**User Story 57: Exact-Phrase Searches**
Owner: Daniel Benes-Magana | Total Hours: 6 hrs
Description: As a user, I would like to perform exact-phrase searches.

| Task | Estimated Hours |
|---|---|
| Add exact-phrase query parser to backend | 2 hrs |
| Update UI search bar to support quote syntax | 2 hrs |
| Integration testing | 2 hrs |

**User Story 96: Manual Corrections Reflected in Search Results**
Owner: Daniel Benes-Magana | Total Hours: 5 hrs
Description: As a user, I would like my manual corrections to be reflected in search results.

| Task | Estimated Hours |
|---|---|
| Update search index and text queries to match against corrected text | 2 hrs |
| Update search hit snippet generator to display corrected text with indicator | 1.5 hrs |
| Integration testing for searching edited and reverted documents | 1.5 hrs |

Acceptance Criteria:
- Given a user has manually corrected text in a document, when a search query matches the corrected text, then the search results include the matching document and page.
- Given a user searches for text that was present in the original OCR output but replaced by a correction, when the search executes, then the obsolete original text is no longer returned as an active match.
- Given search snippets are displayed for a corrected region, when the snippet is rendered, then it accurately reflects the user's updated text.

**User Story 58: Search Extracted Table Contents**
Owner: Jaime Xu | Total Hours: 7 hrs
Description: As a user, I would like to search for extracted table contents.

| Task | Estimated Hours |
|---|---|
| Index structured table text in search database | 3 hrs |
| Update search API to return table cell matches | 2 hrs |

| Task | Estimated Hours |
|---|---|
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given a document contains extracted tables, when I search for text inside a table cell, then the search results include the matching document and snippet.
- Given a search result originates from a table cell, when I click the result, then the viewer navigates to the table and highlights the matching cell.

Given a user searches across multiple documents, when search results are returned, then table cell matches are distinguished with table row and column metadata.

**User Story 59: Sort Search Results**
Owner: Aagam Bapna | Total Hours: 5 hrs
Description:As a user, I want to sort search results so I can find the right document faster.

| Task | Estimated Hours |
|---|---|
| Add sort options to the search API (date, name) | 2 hrs |
| Add a sort dropdown on the search page | 1 hrs |
| Keep the chosen sort when results reload | 0.5 hrs |
| Unit and integration testing | 1.5 hrs |

Acceptance Criteria:

- Given I have search results, when I sort by date, then the list is ordered by upload date.
- Given I have search results, when I sort by name, then the list is alphabetical.
- Given I change the sort, when results load again, then the new order is used.

**User Story 93: OCR Error Tolerance**
Owner: Daniel Benes-Magana | Total Hours: 7 hrs
Description: As a user, I would like search to tolerate common OCR errors (such as 1 vs l, 0 vs O).

| Task | Estimated Hours |
|---|---|
| Implement fuzzy search and character normalization logic | 3 hrs |
| UI display adjustments for fuzzy match highlighting | 2 hrs |
| Integration testing | 2 hrs |

## Administration Enhancements

**Task 71: Disable User Account**
Owner: Cheng-En Lee | Total Hours: 6 hrs
Description: As an administrator, I would like to disable a user's account so they cannot log in, while keeping their documents intact.

| Task | Estimated Hours |
|---|---|
| Add account status flag and disable logic to backend | 2 hrs |
| Add disable account toggle/action in admin UI | 2 hrs |
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given the admin logged in to its account, when the admin disabled a user account, the user should not be able to log in.
- Given the admin logged in to its account and a user's account is disabled, when the admin enabled a user account, the user should be able to log in again.
- Given the user with no admin privileges, the user should not be able to disable other users.

**Task 72: Promote User to Administrator**

Owner: Cheng-En Lee | Total Hours: 4 hrs

Description: As an administrator, I would like to promote a user to administrator.

| Task | Estimated Hours |
|---|---|
| Update user role management API and UI action | 2 hrs |
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given the user with admin privileges, when the user logs in, the user should be able to promote other users to administrators.
- Given the user with no admin privileges, the user should not have access to any admin features.
- Given a user has been promoted to administrator, when they log in, then they are granted access to administrative features and management controls.

**User Story 74: View Processing Jobs Across Users**

Owner: Aagam Bapna | Total Hours: 5 hrs

Description: As an administrator, I would like to view processing jobs across all users, including queue position, current stage, and failure reason.

| Task | Estimated Hours |
|---|---|
| Create backend API to fetch system-wide job queue status | 2 hrs |
| Build job monitoring table view in admin UI | 2 hrs |
| Integration testing | 1 hrs |

Acceptance Criteria:

- Given I am an admin, when I open the jobs page, then I see jobs from every user.
- Given a job is running, when I view it, then I see its queue place and current stage.
- Given a job failed, when I view it, then I see why it failed.

**Task 76: Worker Node and Server Status**

Owner: Jaime Xu | Total Hours: 6 hrs

Description: As an administrator, I would like to see the status of each worker node and inference server.

| Task | Estimated Hours |
|---|---|
| Implement health-check endpoint for worker nodes and servers | 2 hrs |
| Display worker node status dashboard in admin panel | 2 hrs |
| Integration testing | 2 hrs |

Acceptance Criteria:

- Given an administrator visits the admin dashboard, when worker nodes and inference servers are operational, then each node's health status, hostname, and response latency are shown.
- Given a worker node or inference server becomes unreachable or unhealthy, when the health check executes, then the admin dashboard marks the service as degraded/down and highlights the issue.
- Given a non-administrator user requests the node status endpoint, then the server rejects the request with a 403 Forbidden error.

**User Story 30: View Processing Job Progress**

Owner: Jaime Xu | Total Hours: 4 hrs

Description: As a user, I would like to view the progress of a document-processing job while it is running.

| Task | Estimated Hours |
|---|---|
| Implement real-time processing status tracking API endpoint | 2 hrs |
| Build progress bar indicator UI component in document view | 1 hr |
| Integration testing | 1 hr |

Acceptance Criteria:

- Given a document is actively being processed, when I open its view, then I see a progress bar indicating the percentage of completed pages.
- Given the worker completes an individual page, when status is polled or streamed, then the progress indicator increments without refreshing the whole page.
- Given processing reaches 100%, when the job completes, then the viewer smoothly transitions to displaying the finished extracted content.

**Task 83: See Processing Status for Each Page**

Owner: Jaime Xu | Total Hours: 5 hrs

Description: As a user, I would like to see processing status for each page, not only the document as a whole.

| Task | Estimated Hours |
|---|---|
| Implement incremental page-level status updates in worker nodes | 2 hrs |
| Add page status indicators and progress badges to viewer UI | 2 hrs |
| Integration testing | 1 hr |

Acceptance Criteria:

- Given a multi-page document is being parsed, when the worker completes or fails an individual page, then the worker commits that page's status immediately.

- Given some pages succeed while others fail, when I view the document, then each page clearly shows a distinct status badge (queued, processing, completed, or failed).
- Given a page fails during parsing, when I inspect the failed page in the viewer, then a page-specific error banner is rendered explaining the failure reason.

**Task 85: Record Model Version Produced**
Owner: Jaime Xu | Total Hours: 4 hrs
Description: As a user, I would like each result to record which model version produced it so I can reprocess when models improve.

| Task | Estimated Hours |
|---|---|
| Add model version metadata to backend OCR pipeline and schema | 2 hrs |
| Display model version information in document details view | 1 hr |
| Integration testing | 1 hr |

Acceptance Criteria:
- Given a document is parsed by the OCR pipeline, when results are saved, then the inference model name and version are stored alongside the document metadata.
- Given I inspect a processed document's details, when viewing its metadata, then the model version that generated the extraction is clearly displayed.
- Given a document is reprocessed with an updated model, when reprocessing finishes, then the document details reflect the new model version.

**Task 99: Keyboard Shortcuts in the Viewer**
Owner: Jaime Xu | Total Hours: 3 hrs
Description: As a user, I would like keyboard shortcuts in the viewer (next page, next low-confidence region, toggle boxes).

| Task | Estimated Hours |
|---|---|
| Implement keyboard shortcut listeners for viewer navigation and toggles | 1.5 hrs |
| Add keyboard shortcut cheat sheet help modal in viewer UI | 1 hr |
| Frontend testing | 0.5 hrs |

Acceptance Criteria:
- Given I am viewing a document, when I press navigation keys (e.g., Left/Right arrows or J/K), then the viewer navigates to the previous or next page.
- Given bounding boxes are visible, when I press the toggle shortcut (e.g., 'B'), then bounding box overlays toggle between visible and hidden.
- Given I press the shortcut help key (e.g., '?'), when the viewer is active, then an accessible modal dialog listing all keyboard shortcuts appears.

**Task 98: Page Thumbnails for Navigating Long Documents**
Owner: Jaime Xu | Total Hours: 4 hrs
Description: As a user, I would like page thumbnails for navigating long documents.

| Task | Estimated Hours |
|---|---|
| Generate and cache page preview thumbnails in worker/storage pipeline | 1.5 hrs |
| Build collapsible thumbnail sidebar navigation in viewer | 1.5 hrs |
| Integration testing and active page synchronization | 1 hr |

Acceptance Criteria:

- Given a multi-page document is uploaded, when document pages are rendered, then low-resolution thumbnail images are generated and stored.
- Given I open a multi-page document in the viewer, when the thumbnail sidebar is expanded, then a scrolling list of numbered page thumbnails is displayed.
- Given I click a page thumbnail, when selected, then the document pane immediately scrolls to that page and updates the active page indicator.

**Task 75: Force-Cancel and Retry Failed Jobs System-Wide**

Owner: Jaime Xu | Total Hours: 4 hrs

Description: As an administrator, I would like to force-cancel a stuck job and retry a failed job system-wide.

| Task | Estimated Hours |
|---|---|
| Implement admin endpoints to force-cancel stuck jobs and retry failed jobs | 2 hrs |
| Add cancel and retry action buttons with confirmation modals to admin jobs UI | 1 hr |
| Integration testing for admin permissions and job state transitions | 1 hr |

Acceptance Criteria:

- Given a processing job is stuck across any user's library, when an administrator triggers force-cancel, then the job is revoked, marked canceled, and worker resources are released.
- Given a processing job has failed, when an administrator triggers retry, then a new job is queued for the document without requiring the user to re-upload.
- Given a non-administrator user attempts to invoke the admin force-cancel or retry endpoint for another user's job, then the request is forbidden with a 403 error.

**Task 82: Storage Quota Usage Indicator**

Owner: Jaime Xu | Total Hours: 3 hrs

Description: As a user, I would like to see how much of my storage quota I have used.

| Task | Estimated Hours |
|---|---|
| Build user storage quota indicator and progress bar UI component | 1.5 hrs |
| Add near-capacity warning banners and threshold alerts in dashboard and settings | 1 hr |
| Integration and accessibility testing | 0.5 hrs |

Acceptance Criteria:

- Given I am signed in, when I navigate to the dashboard or settings, then I see my storage quota usage displayed as both a formatted size (e.g., "120 MB of 1 GB used") and a visual progress bar.

- Given my storage usage exceeds 80% or 95% of quota, when I view the uploader or dashboard, then a warning banner informs me that my storage is nearly full.

Given I upload or delete documents, when storage usage changes, then the quota indicator updates to reflect the latest storage consumed.

**User Story 102: Admin Delete User Account**

Owner: Aagam Bapna | Total Hours: 5 hrs

Description:As an administrator, I want to delete a user account so they cannot log in and they no longer show in the admin table.

| **Task** | **Estimated Hours** |
|---|---|
| Add an admin API to delete a user | 1.5 hrs |
| Add a delete action on the admin accounts table | 1.5 hrs |
| Remove the account so login no longer works | 1 hr |
| Unit and integration testing | 1 hr |

Acceptance Criteria:

- Given I am an admin, when I delete a user, then that user cannot log in.
- Given I delete a user, when I open the accounts table, then that row is gone.
- Given I am not an admin, when I try to delete a user, then the request is rejected.
- Given the user has documents, they are deleted permanently from the system

## Document Library

**User Story 101: Soft-Delete and Restore Documents**

Owner: Aagam Bapna | Total Hours: 5 hrs

Description: As a user, I want deleted documents to stay recoverable for a short time so I can undo a mistake.

| **Task** | **Estimated Hours** |
|---|---|
| Soft-delete a document instead of wiping it right away | 1.5 hrs |
| Restore a document within the time window | 1.5 hrs |
| Permanently delete after the window ends and free storage | 1 hr |
| Unit and integration testing | 1 hr |

Acceptance Criteria:

- Given I delete a document, when I am still inside the time window, then I can restore it.
- Given I restore a document, when it comes back, then its extracted content is still there.

- Given the time window (set by user) has ended, when cleanup runs, then the document is gone and storage is freed.

**User Story 103: Folders in My Library**

Owner: Aagam Bapna | Total Hours: 5 hrs
Description: As a user, I want to put documents in folders so I can keep my library organized.

| Task | Estimated Hours |
|---|---|
| Create, rename, and delete folders | 2 hrs |
| Move a document into or out of a folder | 1.5 hrs |
| Show the library by folder | 0.5 hr |
| Unit and integration testing | 1 hr |

Acceptance Criteria:

- Given I create a folder, when I move a document into it, then that document shows in that folder and not in the main list.
- Given I rename a folder, when I open it, then the documents are still inside.
- Given I delete a folder, when I confirm, then the folder is gone and its documents go back to the main library (they are not wiped).

**User Story 105: Tag Documents**

Owner: Aagam Bapna | Total Hours: 5 hrs
Description: As a user, I want to add tags to documents so I can find groups of files without using folders.

| Task | Estimated Hours |
|---|---|
| Create, rename, and delete tags | 1.5 hrs |
| Add or remove tags on a document | 1.5 hrs |
| Filter the library by tag | 1 hr |
| Unit and integration testing | 1 hr |

Acceptance Criteria:

- Given I add a tag to a document, when I filter by that tag, then only documents with that tag are shown.
- Given I remove a tag from a document, when I filter by that tag, then that document is gone from the list.
- Given I delete a tag, when I look at my library, then the tag is gone and the documents themselves are still there.

# Backlog

**Functional Requirements**

| Task # | Functional Requirement | Estimated Hours on Task |
|---|---|---|
| 1 | As a user, I would like to create an account | 2 |
| 2 | As a user, I would like to login to the account I have created | 1 |
| 3 | As a user, I would like to view all the documents associated with my account | 3 |
| 4 | As a user, I would like to edit my login credentials | 1 |
| 5 | As a user, I would like to delete documents associated with my account | 1 |
| 6 | As a user, I would like to delete my account | 1 |
| 7 | As a user, I would like to upload a PDF document so that its contents can be processed. | 2 |
| 8 | As a user, I would like to upload multiple documents at once so that I do not need to submit each file individually. | 2 |
| 9 | As a user, I would like to process the documents I have uploaded by parsing their text content. | 4 |
| 10 | As a user, I want temporary worker or processing failures to not result in loss of my uploaded document or previously completed processing data | 5 |
| 11 | As a user, I would like to know when document processing has failed. | 2 |
| 12 | As a user, I want to view processed documents individually and their extracted text content | 3 |
| 13 | As a user, I want to view the bounding boxes on each processed document that links to certain extracted text content. | 2 |
| 14 | As a user, I would like to look up specific pages of processed documents by text search. | 4 |
| 15 | As a user, I would like to perform advanced searches across documents using filters like date ranges or document type. | 3 |
| 16 | As a user, I would like to export extracted content for external use, such as markdown | 3 |

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| 17 | As a user, I would like to connect my cloud storage (e.g., Google Drive/Dropbox) to automatically import documents. | 5 |
| 18 | As a user, I would like to share processed documents and their bounding-box data with other users for collaborative review. | 8 |
| 19 | As a user, I would like to securely log out of my account. | 2 |
| 20 | As a user, I would like the system to preserve the original page number associated with each piece of extracted information | 4 |
| 21 | As a user, I would like extracted document information to be represented in structured JSON so that it can be used programmatically | 6 |
| 22 | As a user, I would like the system to identify different document regions such as headers, paragraphs, figures, and tables. | 8 |
| 23 | As a user, I would like tables to be extracted while preserving their rows, columns, and structure. | 8 |
| 24 | As a user, I would like tables that continue across page breaks to be reconstructed as one logical table. | 10 |
| 25 | As a user, I would like to view the original document and its extracted structured data side-by-side. | 6 |
| 26 | As a user, I would like to click extracted data and have the corresponding location highlighted on the original document. | 6 |
| 27 | As a user, I would like search results to show the document, page, and matching extracted content where the result originated. | 5 |
| 28 | As a user, I would like to click a search result and navigate directly to the corresponding location in the source document. | 5 |
| 29 | As a user, I would like failed document-processing jobs to be retried without uploading the document again. | 5 |
| 30 | As a user, I would like to view the progress of a document-processing job while it is running. | 4 |
| 31 | As a user, I would like to view the overall progress of a batch containing multiple documents. | 5 |
| 32 | As a user, I would like to receive warnings that my document may be blurred instead of silently receiving poor extraction results. | 3 |
| 33 | As a user, I would like to receive warnings that my | 3 |

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| | document may be skewed instead of silently receiving poor extraction results. | |
| 34 | As a user, I would like to receive warnings that my document may be cropped/truncated instead of silently receiving poor extraction results. | 3 |
| 35 | As a user, I would like to receive warnings that my document may be too dark/too light instead of silently receiving poor extraction results. | 3 |
| 36 | As a user, I would like to upload PNG and JPEG document scans in addition to PDFs. | 5 |
| 37 | As a user, I would like unsupported file types to be rejected with an explanation. | 3 |
| 38 | As a user, I would like to view document metadata including file name, page count, upload date, and processing status. | 4 |
| 39 | As a user, I would like to rename uploaded documents. | 3 |
| 40 | As a user, I would like to sort my documents by name, upload date, or status. | 3 |
| 41 | As a user, I would like to filter my document library by processing status. | 3 |
| 42 | As a user, I would like to cancel a queued document-processing job. | 5 |
| 43 | As a user, I would like to reprocess a document without uploading it again. | 4 |
| 44 | As a user, I would like the system to preserve the correct reading order in multi-column documents. | 10 |
| 45 | As a user, I would like rotated pages to be detected and corrected before OCR. | 8 |
| 46 | As a user, I would like extracted text to include a confidence score. | 6 |
| 47 | As a user, I would like low-confidence extracted regions to be visually marked for review. | 5 |
| 48 | As a user, I would like figure captions to remain associated with their figures. | 8 |
| 49 | As a user, I would like extracted table data to be exportable as CSV. | 5 |
| 50 | As a user, I would like extracted data to be exportable as JSON. | 4 |
| 51 | As a user, I would like to manually correct incorrectly extracted text. | 8 |

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| 52 | As a user, I would like my corrections to be saved. | 5 |
| 53 | As a user, I would like to toggle bounding boxes on and off. | 2 |
| 54 | As a user, I would like bounding boxes to remain aligned while zooming the document. | 6 |
| 55 | As a user, I would like search matches highlighted after navigating to their source page. | 4 |
| 56 | As a user, I would like search results to include a short surrounding text snippet. | 4 |
| 57 | As a user, I would like to perform exact-phrase searches. | 4 |
| 58 | As a user, I would like to search for extracted table contents. | 6 |
| 59 | As a user, I would like to sort search results. | 3 |
| 60 | As a user, I would like to receive a notification when a long-running batch completes. | 4 |
| 61 | As a developer, I would like automated unit tests for critical backend API functionality. | 10 |
| 62 | As a developer, I would like automated tests for OCR and document-processing functionality. | 10 |
| 63 | As a developer, I would like integration tests covering document upload, processing, storage, and retrieval. | 12 |
| 64 | As a developer, I would like integration tests for table and layout extraction using known test documents. | 12 |
| 65 | As a developer, I would like end-to-end tests covering major user workflows. | 12 |
| 66 | As a developer, I would like tests to run automatically when code changes are merged. | 12 |
| 67 | As a user, I would like to reset my password if I forget it. | 5 |
| 68 | As a user, I would like to search my document library by document name | 4 |
| 69 | As a user, I would like to view information about why a document-processing job failed so that I can determine how to correct the problem. | 5 |
| 70 | As an administrator, I would like to view a list of all accounts with name, email, join date, and document count | 2 |
| 71 | As an administrator, I would like to disable a user's | 2 |

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| | account so they cannot log in, while keeping their documents intact. | |
| 72 | As an administrator, I would like to promote a user to administrator | 1 |
| 73 | As an administrator, I would like a dashboard showing total users, documents, storage used, and jobs by state. | 3 |
| 74 | As an administrator, I would like to view processing jobs across all users, including queue position, current stage, and failure reason | 5 |
| 75 | As an administrator, I would like to force-cancel a stuck job and retry a failed job system-wide | 3 |
| 76 | As an administrator, I would like to see the status of each worker node and inference server | 3 |
| 77 | As a user, I would like PDFs that already contain a text layer to skip OCR and reuse the embedded text and positions. | 6 |
| 78 | As a user, I would like to be prompted for a password when uploading an encrypted PDF instead of the job failing silently. | 3 |
| 79 | As a user, I would like to be warned when I upload a document identical to one I already have. | 4 |
| 80 | As a user, I would like to see upload progress per file and retry a failed large upload without starting over. | 5 |
| 81 | As a user, I would like to upload documents by dragging and dropping files or a folder. | 3 |
| 82 | As a user, I would like to see how much of my storage quota I have used. | 3 |
| 83 | As a user, I would like to see processing status for each page, not only the document as a whole. | 5 |
| 84 | As a user, I would like only failed pages to be retried instead of the whole document. | 6 |
| 85 | As a user, I would like each result to record which model version produced it so I can reprocess when models improve. | 4 |
| 86 | As a user, I would like to be warned when a page contains handwriting so I know extraction quality may be lower. | 5 |
| 87 | As a user, I would like the system to detect document language and use the appropriate OCR model. | 6 |
| 88 | As a user, I would like repeated headers, footers, | 8 |

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| | and page numbers to be detected and excluded from paragraphs and tables. | |
| 89 | As a user, I would like footnotes to be linked to the text they reference. | 6 |
| 90 | As a user, I would like figures to be extracted as image files and included in exports. | 5 |
| 91 | As a user, I would like each table cell to have its own bounding box. | 6 |
| 92 | As a user, I would like small jobs to be prioritized so they are not stuck behind a large batch. | 5 |
| 93 | As a user, I would like search to tolerate common OCR errors (such as 1 vs l, 0 vs O). | 6 |
| 94 | As a user, I would like to search within a single document. | 3 |
| 95 | As a user, I would like to filter search results by confidence score. | 3 |
| 96 | As a user, I would like my manual corrections to be reflected in search results. | 5 |
| 97 | As a user, I would like to search table contents by column header. | 6 |
| 98 | As a user, I would like page thumbnails for navigating long documents. | 4 |
| 99 | As a user, I would like keyboard shortcuts in the viewer (next page, next low-confidence region, toggle boxes). | 3 |
| 100 | As a user, I would like to see the difference between the original extraction and my correction, and undo a correction. | 6 |
| 101 | Soft-delete a document, restore it within N time, then permanent delete. | 5 |
| 102 | Admin can delete an accounts login and remove it from the admin table. | 5 |
| 103 | As a user, I want to put documents in folders so I can keep my library organized. | 5 |
| 105 | As a user, I want to add tags to documents so I can find groups of files without using folders. | 5 |

**Non-Functional Requirements**

| Task # | Non-Functional Requirement | Estimated Hours on Task |
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| 1 | As a user, I would like to set up two-factor authentication for my account to increase security. | 3 |
| 2 | As an administrator, I would like to scale the document processing service by running more worker nodes and inference servers | 5 |
| 3 | As a user, I want the application to have an intuitive web interface supporting desktop, laptop, and mobile screens. | 5 |
| 4 | As a user, I want the application to be responsive. This means the application should provide appropriate loading indicators or allow me to continue using other parts of the application without interruption. | 5 |
| 5 | As a user, I expect uploaded documents and their extracted contents to be only accessible by authorized users | 3 |
| 6 | As a user, I expect my authentication credentials to be securely stored and transmitted | 4 |
| 7 | As a user, I want search results across processed documents to be returned within a reasonable response time under normal system load | 4 |
| 8 | As a user, I want long-running OCR and document-processing jobs to run in the background so that I can continue using other parts of the application | 5 |
| 9 | As a user, I want the failure of one document processing job to not cause other documents in the batch to fail | 4 |
| 10 | As a user, I want failed processing jobs to be recoverable or retryable without needing to upload the document again | 4 |
| 11 | As an administrator, I want additional worker nodes to be added without redesigning the document processing system so that processing capacity can scale | 5 |
| 12 | As a user,  I want clear error | 4 |

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| | messages that explain when an upload, processing job, or search operation fails | |
| 13 | As a user, I want document processing states such as queued processing, completed, and failed to be clearly distinguishable | 4 |
| 14 | As a developer, I want the document processing, search, storage, and interface components to be sufficiently modular so that they can be tested and modified independently | 5 |
| 15 | As a user, I want the application to support current versions of major desktop browsers so that I can access the system reliably across common environments. | 4 |
| 16 | As a user, I want core application workflows to remain usable through keyboard navigation and clearly labeled interface elements. | 5 |
| 17 | As a user, I expect deleting a document to also remove its associated extracted data and search-indexed content. | 5 |
| 18 | As a developer, I want processing failures to generate logs that identify the affected document, job, and processing stage so that problems can be diagnosed efficiently. | 6 |
| 19 | As a developer, I want environment-specific configuration such as credentials, storage locations, and inference endpoints to be configurable rather than hard-coded. | 5 |
| 20 | As a user, I want previously completed document-processing results to remain available even if a worker or inference service temporarily fails. | 5 |
| 21 | As an administrator, I would like to set global limits such as maximum file size and maximum documents per user, so one account can't consume all storage. | 3 |
| 22 | As an administrator, I would like a log of administrative actions, | 3 |

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| | recording who disabled or deleted what and when | |

## Developer Hours Summary

| **Developer** | **Assigned Hours** |
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| Aagam Bapna | 40 hrs |
| Anthony McCrovitz | 40 hrs |
| Cheng-En Lee | 38 hrs |
| Daniel Benes-Magana | 40 hrs |
| Jaime Xu | 40 hrs |